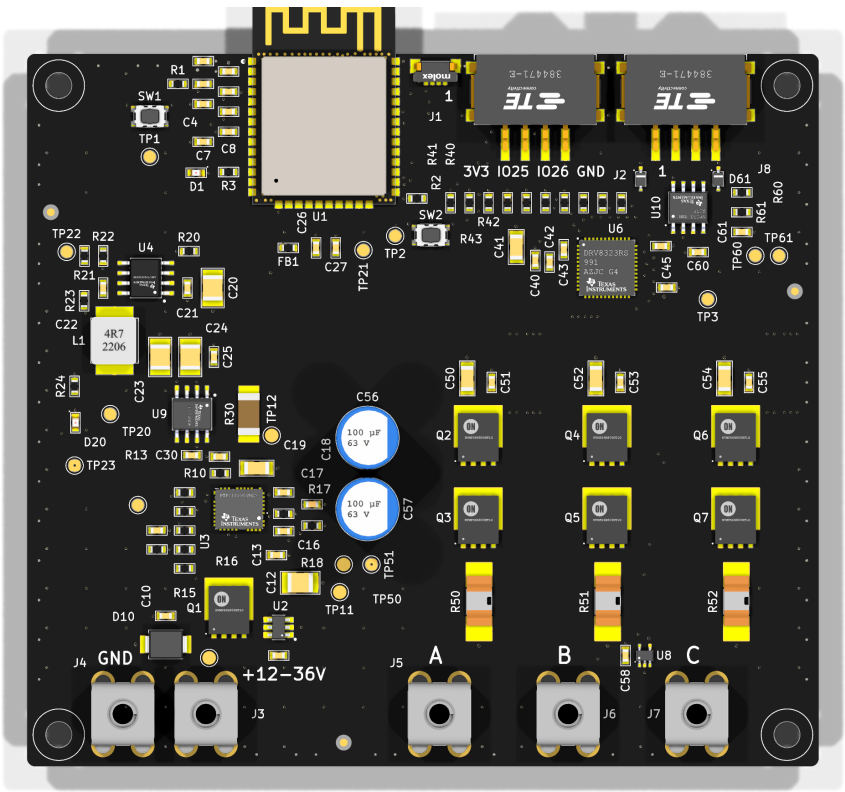


ESC3Phase Rev B

3-phase BLDC/PMSM ESC · ESP32-WROOM-32E · 12–36 V · 5–10 A



Project	PCBDesignWClaude, KiCad project <i>hardware/ESC3Phase</i> , commit 2d62300 (main)
Board	100 × 90 mm, 4 layers, 1.6 mm FR-4, 1 oz all layers, LF-HASL, 109 parts (102 SMD, all top)
Rules applied	METU PowerLab PCB_DESIGN_RULES (PCBWay standard spec) and the schematic's own LAYOUT: notes
Tools	kicad-cli 10.0.5 (ERC, DRC with zone refill + schematic parity), KiCad pcbnew Python API, datasheet math
Date	2026-10-05

1 Summary

Rev B passes every automated check: kicad-cli reports **ERC 0 errors** and **DRC 0 errors, 0 unconnected items and 0 schematic-parity issues**, reproduced today with zones refilled. The schematic is clean and well documented, and the Rev A net shorts are gone. The part choices are checked against their datasheets.

Passing DRC does not mean the board will work at its rated current. This review looked at what DRC cannot see: current capacity, return paths, protection against a rising bus voltage, measurement range, thermal design and assembly. It found **2 high**, **7 medium** and **8 low** findings. None of them stops a first prototype being built and powered on a current-limited bench supply. The two high findings need attention before the board runs a real motor at 10 A or brakes one:

- **H1. Nothing limits the bus voltage when the motor feeds energy back.** The LM74700 ideal diode stops current from flowing back into the supply, and the only TVS (D10) sits before it on the input. When the motor brakes or slows down, the energy it returns can only charge the 200 μ F on the bus, and the DRV8323 is destroyed above 65 V.
- **H2. All high-current copper is on L1 only.** The project's own LAYOUT notes ask for "pours on both layers with via arrays (5–10 A)", but the bus, phase and input pours have no copper on any other layer. Each low-side shunt sends the full phase current to ground through only 1–2 vias next to its pad. The eFuse input and output necks are 2.4–2.8 mm wide.

Area	High	Medium	Low	Status
Automated checks (ERC / DRC / parity)	–	–	1	Pass; library pin-type warnings, Fab-only references, rails
Protection & power path	1	2	–	Bus over-voltage on braking; eFuse surge on EN/UVLO; preproduction eFuse part
Copper, current & thermal	1	2	3	Single-layer power copper, few return vias, heat path, bulk caps, TMP235
Sensing & control	–	2	3	No bus-voltage sense; low end of the INA240 range unreadable; no Hall/encoder input
Assembly / DFM	–	1	–	Thermal vias open on both sides; bare bottom copper
Documentation	–	–	1	Root sheet still says 2 layers

Verdict: fine to order as a prototype for bring-up, firmware work and low-current motor tests. Fix H1 and H2 (and preferably M1, M2 and M4) in a Rev C before using the 5–10 A rating. Until then, firmware should limit braking current and the board should run from a supply that can absorb energy (a battery) or behind a clamp.

2 Verification results (reproduced 2026-10-05)

Check	Command	Result
ERC	kiCad-cli sch erc --severity-all	0 errors , 11 warnings: all <i>pin_to_pin</i> Bidirectional ↔ Power output (connector pins typed bidirectional on nets with a PWR_FLAG or DRV8323 pin 45)
DRC	kiCad-cli pcb drc --refill-zones --schematic-parity --severity-all	0 errors , 2 warnings: U1 (ESP32) silk outline clipped by the board edge and mask (antenna overhang, intentional)
Unconnected	same run	0
Schematic parity	same run	0 missing or extra parts or nets
Track corners	tools/pcb/angle_check.py (from the project)	no corner sharper than 45° (as documented)
Fab outputs	hardware/fab	Gerber/drill zip, BOM (UTF-8), position file (102 SMD), assembly PDFs present

Board statistics from the KiCad Python API

Item	Value	Item	Value
Outline	100.1 × 90.1 mm, R1 corners	Copper layers	4 (L2 solid GND, L1/L3/L4 GND-filled)
Vias	330 (311 × 0.3/0.6 mm, 19 × 0.4/0.8 mm)	GND vias	218
Holes	445 PTH, 8 NPTH (4 × M3 + 4)	Min track	0.15 mm (one neck-down), 0.2 mm elsewhere
L1 GND fill	4183 mm², 39 islands	L2 / L3 / L4 GND	8277 / 7595 / 7429 mm²
+VBUS_PROT pour	717 mm², L1 only, 0 vias	PHASE_A/B/C pours	≈ 311 mm² each, L1 only

Design overview

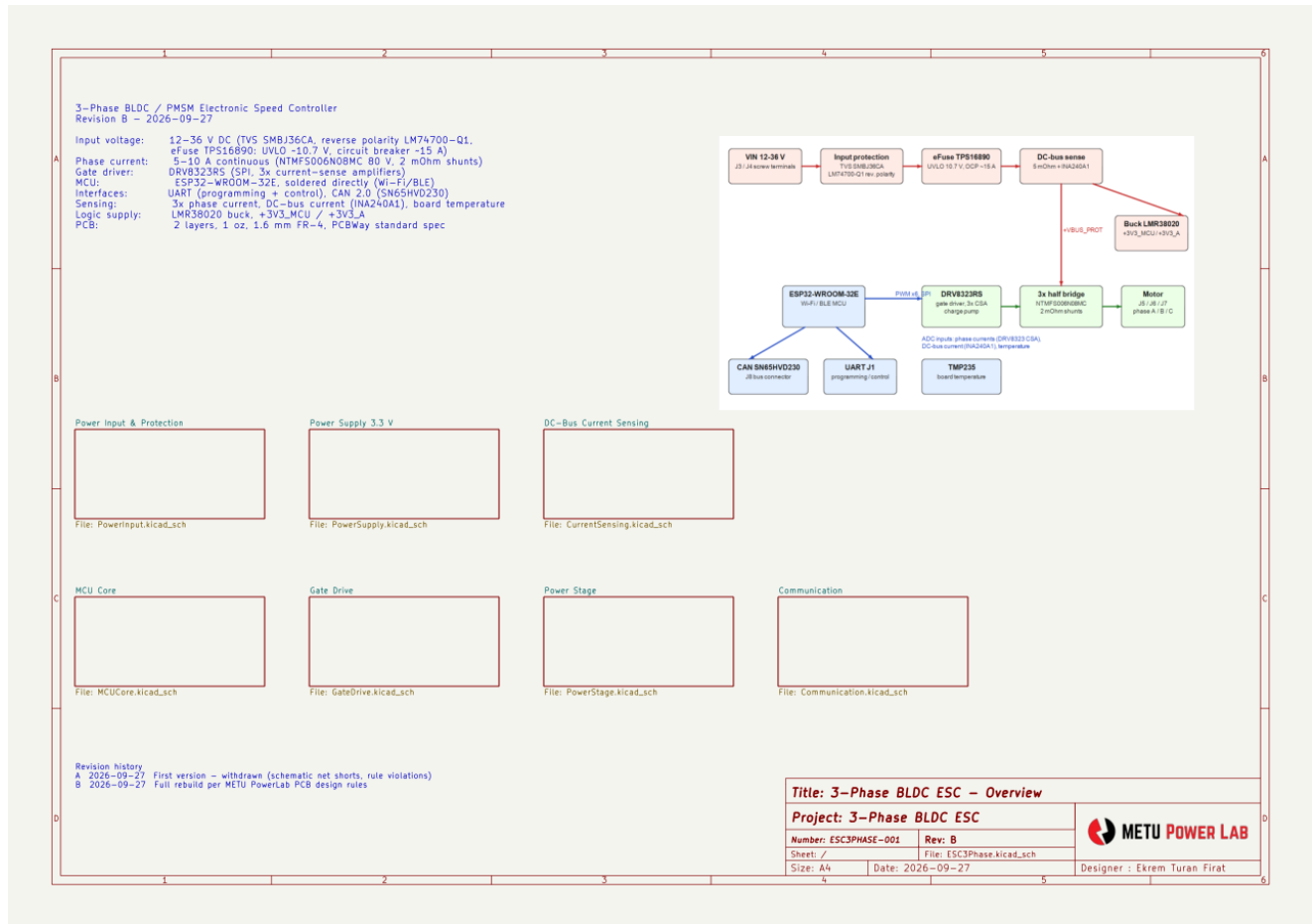


Figure 1 — Root sheet (schematic page 1): specs, block diagram and the sheet symbols.

Power flows from J3 (VIN_RAW) through the TVS D10, the reverse-polarity FET Q1 with the LM74700-Q1 and the TPS16890 eFuse, then through the 5 mΩ DC shunt R30, into +VBUS_PROT. +VBUS_PROT feeds three NTMFS006N08MC half bridges (each with a 2 mΩ low-side shunt), the DRV8323RS and the LMR38020 3.3 V buck. The ESP32 drives 6 PWM lines plus SPI, and reads the three DRV8323 current-amplifier outputs, the INA240A1 (DC current) and the TMP235 on ADC1.

3 Findings at a glance

ID	Severity	Finding	Area
H1	High	No over-voltage clamp or bus-voltage limit on +VBUS_PROT; the ideal diode blocks energy returned by braking	Protection
H2	High	High-current copper on L1 only; 1–2 return vias per shunt; 2.4–2.8 mm eFuse necks (LAYOUT notes not met)	Copper
M1	Medium	No bus-voltage measurement for the MCU; all ADC1 inputs already used	Sensing
M2	Medium	INA240 output (REF = GND) is below the ESP32 ADC floor up to about 1–1.5 A	Sensing
M3	Medium	U3 is ordered as PTPS16890VMAR (P = preproduction), with no distributor part numbers	BOM
M4	Medium	Thermal vias are exposed on the bottom (MOSFETs 0.5 mm, QFN/ESP32/eFuse pads): solder wicking, bare switching copper	DFM
M5	Medium	Heat path at 10 A not verified: drain pads spread only into L1 pours and isolated bottom pads	Thermal
M6	Medium	EN/UVLO reaches about 7 V (> 6 V abs max) during a 58 V clamped surge (already documented as a residual risk)	Protection
M7	Medium	Bulk caps: ripple-current rating not checked; placed 12–44 mm from the legs, not "between the half bridges"	Power stage
L1	Low	Root sheet says "PCB: 2 layers, 1 oz"; the board is 4 layers	Docs
L2	Low	TMP235 is about 16 mm from the nearest low-side FET, next to the phase terminals	Layout
L3	Low	No gate-resistor or snubber footprints; the DRV8323 default IDRIVE (1 A / 2 A) is aggressive	Gate drive
L4	Low	Hot-loop caps have one 0.3 mm GND via each; the loop closes through L2 over about 33 mm	Layout
L5	Low	No Hall or encoder connector; J2 has only two ADC2 GPIOs, so control is sensorless only	Function
L6	Low	Phase current amplifier gain 20 V/V uses about half the ADC span at 10 A	Firmware
L7	Low	10 µF / 100 V 1206 X6S hot-loop caps lose most of their capacitance at 36 V DC bias	BOM
L8	Low	11 ERC warnings (library pin types); 25 references on the Fab layer only; breakaway rails needed	Housekeeping

Severity: **High**: can damage the board or stop it meeting its rating; fix before 10 A or braking tests. **Medium**: a real performance, reliability or supply risk; fix in Rev C. **Low**: polish, docs or firmware.

4 Detailed findings

H1 High Bus over-voltage when the motor brakes or slows down

Observed. D10 (SMBJ36CA) clamps VIN_RAW, before Q1. Q1 + U2 (LM74700-Q1) is an ideal diode, so it turns off when current reverses, and U3 (TPS16890) sits after it. +VBUS_PROT has no TVS, no brake chopper and no voltage divider to the MCU. The bus capacitance is about 200 μF of polymer plus $5 \times 10 \mu\text{F}$ ceramic (much less at 36 V, see L7).

Why it matters. During braking or a fast speed drop, field-oriented and six-step control both send motor energy back into the bus. With the return path blocked, that energy only charges the bus capacitors: $V = \sqrt{V_0^2 + 2E/C}$. Starting at 36 V, just **0.29 J** lifts 200 μF past the DRV8323's 65 V absolute maximum, and 0.51 J reaches 80 V (the MOSFET rating). A motor with even a modest load stores several joules at speed. A bench supply that cannot sink current makes this worse.

Recommendation. Add a unidirectional TVS on +VBUS_PROT to GND at the bulk caps (an SMCJ-class 36 V part clamps at about 58 V), and add a bus-voltage divider to an ADC1 pin (M1) so firmware can limit braking current. For hard braking, think about a brake-chopper footprint (FET + resistor). Note this limit in the README until Rev C.

H2 High High-current copper is single-layer, with few return vias

Observed. All power pours (VIN_RAW, +VBUS, +VBUS_EFUSE, +VBUS_PROT, PHASE_A/B/C, LS_SRC) exist **only on L1**. Every other layer under them is GND fill, and +VBUS_PROT has **0 vias**. Measured narrowest sections (Figure 2, Figure 4): phase lanes $\approx 5.5 \text{ mm}$, bus band $\approx 7.5 \text{ mm}$, eFuse input neck (+VBUS to U3 pins 22/23) $\approx 2.8 \text{ mm} \times 3 \text{ mm}$, eFuse output strip (pins 7/8 to R30) $\approx 2.4 \text{ mm} \times 4.5 \text{ mm}$. Each low-side shunt (R50–R52) has **one** GND via within 2.2 mm of its GND pad (R51 has two). That pad sits on an L1 GND island between two phase lanes, joined to L2 by about 5 vias spread over 20 mm (Figure 3).

Why it matters. IPC-2221 (outer layer, 1 oz, long conductor) gives: $5.5 \text{ mm} \approx 8.2 \text{ A}$ at $\Delta T = 10^\circ\text{C}$, so about 16°C rise at 10 A and 39°C at 15 A. $2.4 \text{ mm} \approx 4.5 \text{ A}$ and $2.8 \text{ mm} \approx 5.0 \text{ A}$ at 10°C . The necks are short, so the real rise is lower, but they still heat up locally. With space-vector PWM the DC input current reaches about $0.87 \times I_{\text{peak}}$, so about 12 A at 10 A rms. The lab via budget is about 1 A per 0.3 mm via, while the shunt return carries 5–10 A through 1–2 vias nearby and the L1 island. This breaks rule 3.3/3.4 and the Half Bridge and Power Input LAYOUT notes ("pours on both layers with via arrays (5–10 A)", "MOSFET drain pad: copper on both layers", "U3 exposed pad: via array into the +VBUS pour").

Recommendation. Copy the bus band, phase lanes and input chain onto L4 (and L3 where it is free; L2 stays solid GND), and stitch them to L1 with 0.4 mm via arrays placed across the current flow. Put $8\text{--}12 \times 0.4 \text{ mm}$ vias directly at each shunt GND pad and 2–3 vias per hot-loop cap. Widen the U3 input/output necks to $\geq 4 \text{ mm}$ on L1 and copy them to L4. If space runs out, 2 oz outer copper is the alternative, at extra cost (ask first, per rule 2.0).

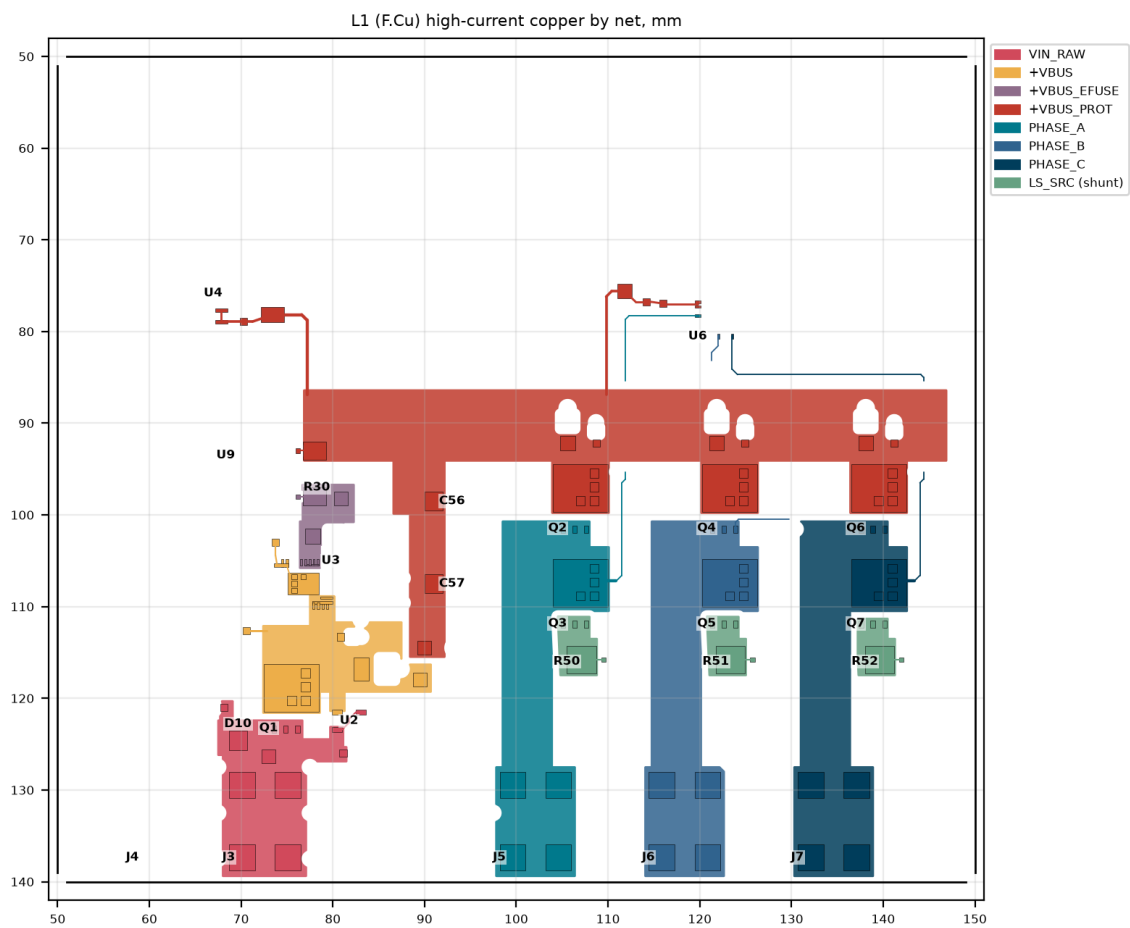
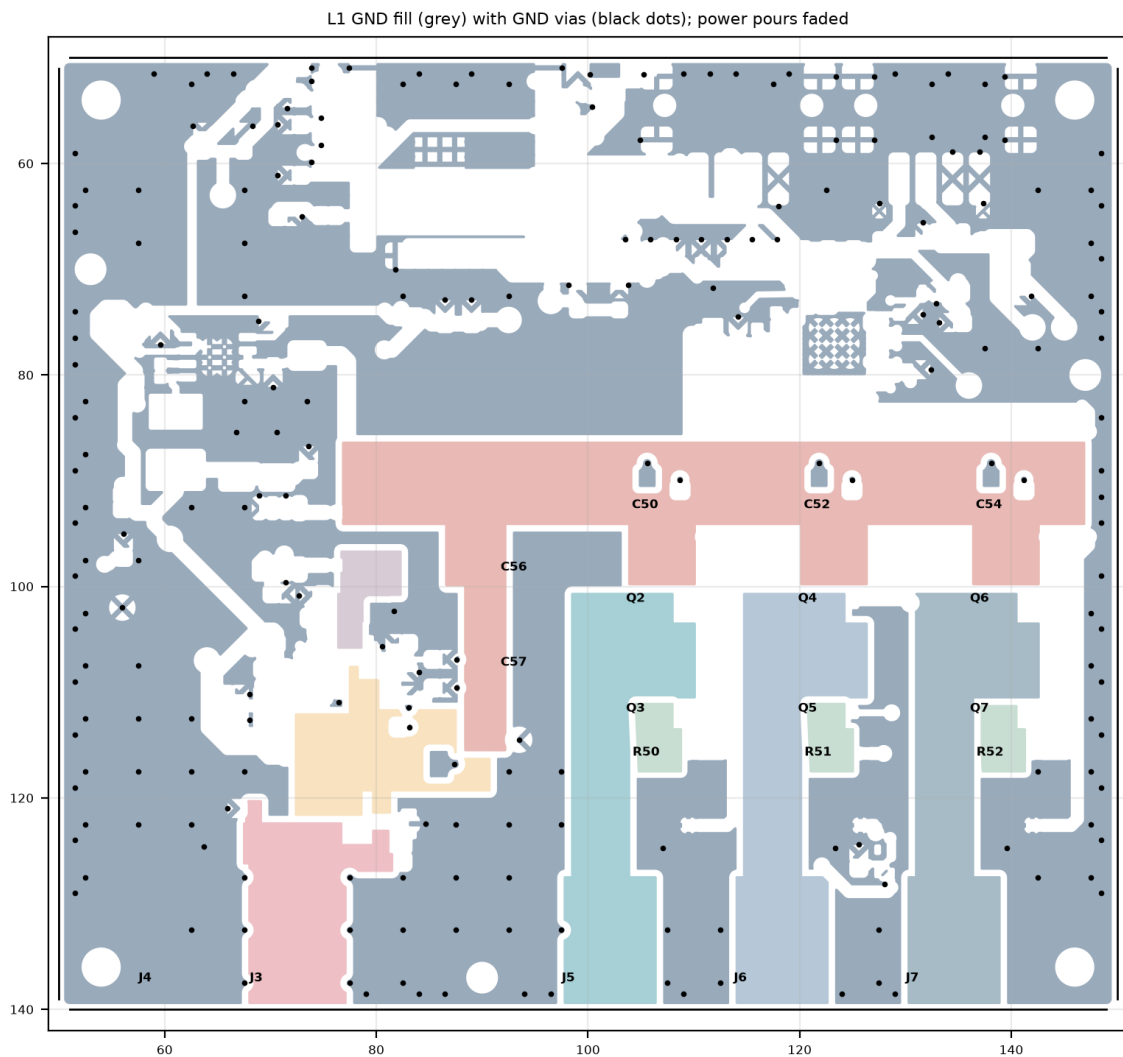


Figure 2 — L1 high-current copper by net (from the board's filled zones). No other layer carries these nets.



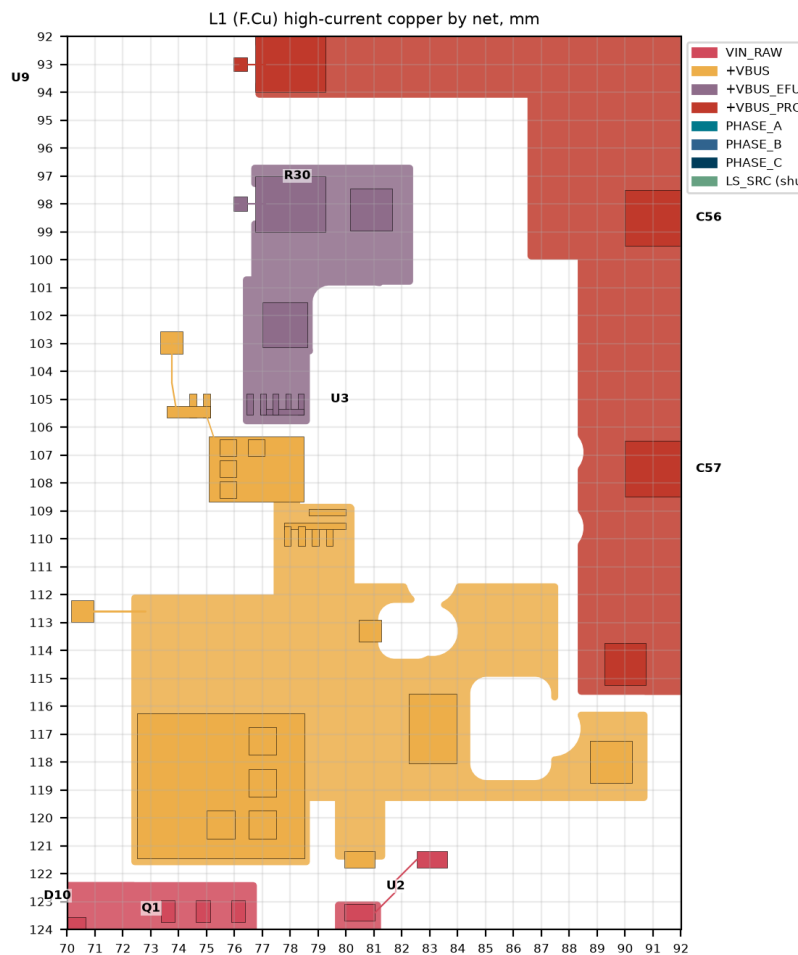


Figure 4 — eFuse area: U3 input neck (+VBUS, orange) and output strip to R30 (+VBUS_EFUSE, purple). 1 mm grid.

M1 Medium No bus-voltage measurement

Observed. The ESP32 pin map has phase currents (IO32/33/34), IBUS (VP), TEMP (VN) and EFUSE_NFLT (IO35) on ADC1. No pin reads +VBUS_PROT. The eFuse I²C pins are only pulled up, and EXP1/EXP2 (IO25/26) are ADC2 pins, which cannot be used while Wi-Fi is on.

Why it matters. Field-oriented control needs the bus voltage to scale its duty cycles, and over- and under-voltage shutdown needs it too, which H1 depends on.

Recommendation. Add a divider of about 1:20 (e.g. 200 k / 10 k + 220 pF) from +VBUS_PROT to IO35, and move EFUSE_NFLT to a digital pin or wire-OR it with NFAULT (both are open-drain).

M2 Medium The low end of the DC-bus current range cannot be read

Observed. U9 INA240A1 ($20 \text{ V/V} \times 5 \text{ m}\Omega = 100 \text{ mV/A}$) has REF1 = REF2 = GND, so its output is unidirectional from 0 V. The ESP32 ADC's specified range starts at 100–150 mV, depending on attenuation.

Why it matters. Bus currents below about 1–1.5 A read as zero, and the output stage of an INA240 driven to the ground rail is less accurate there anyway. This hurts idle, light-load and efficiency measurements.

Recommendation. Shift the output up with a small REF offset (a low-impedance divider on REF1/REF2 per SBOS662, about 0.2–0.3 V), or measure at 0 dB / 2.5 dB attenuation and accept a smaller full scale. Calibrate the offset in firmware.

M3 Medium Preproduction eFuse in the BOM

Observed. U3 is listed as **PTPS16890VMAR**, with no Digi-Key or Mouser number. A "P" prefix on a TI part means a preproduction sample.

Why it matters. PCBWay assembly may not be able to source it. Preproduction silicon or specifications can change before release.

Recommendation. Confirm that the production TPS16890VMAR is available, and update the library MPN and distributor fields. If it is not available, plan a second source or a footprint-compatible option before ordering assembly.

M4 Medium Thermal vias exposed on the bottom

Observed. Q1–Q7 footprints: 9×0.5 mm plated vias in the drain pad with **mask open on both sides**, plus a 6.0×5.2 mm bare copper pad on B.Cu. U1 (ESP32 pad 39), U6 (DRV8323 EP) and U3 (eFuse EP) also have B.Cu pads inside a mask opening that contains their vias (visible as bare squares in the bottom render). ARCHITECTURE.md says these vias are "tented underneath".

Why it matters. During reflow, solder runs from the top pads down open 0.3–0.5 mm vias. This leaves voids under the MOSFET and QFN thermal pads and solder blobs on the bottom. It also leaves bare +VBUS_PROT and switching PHASE copper on the bottom, which can short to a metal plate or standoff.

Recommendation. Remove the B.Mask opening from these bottom pads (tent from below, as the docs intend), or keep it only if a heatsink with an insulating pad is planned, and say so. Use 0.3 mm vias in the MOSFET pads (rule 3.4 thermal vias) to reduce wicking. Fix this upstream in PowerLabKiCadLibraries.

M5 Medium Heat path at 10 A not verified

Observed. The drain vias of each MOSFET connect only to an isolated 31 mm² bottom pad, because the inner layers are GND. Heat therefore spreads mainly through the L1 pours. No heatsink or airflow is specified.

Why it matters. First-order estimate at 10 A rms: conduction $\approx I^2 \cdot R_{DS(on),hot} \approx 0.5\text{--}0.9$ W per leg, plus switching loss ($\approx 0.5 \cdot V \cdot I_{sw} \cdot f$, about 0.1–0.4 W per leg at 20 kHz, set by IDRIVE), plus Q1 ($\approx 0.5\text{--}0.9$ W), the shunts and R30 (≈ 1 W). That is 3–6 W total on a 90 cm² board in still air, so parts may run well above 60 °C.

Recommendation. Make the drain copper bigger and use more layers (H2). Plan a bottom heatsink area with an insulating thermal pad (M4). Measure FET case temperature with a thermocouple at 5 A and 10 A before giving the board a rating.

M6 Medium eFuse EN/UVLO above its absolute maximum during a surge

Observed. R11 866 k / R12 118 k sets UVLO to 10.1 V. During a 58 V clamped surge, EN/UVLO reaches about 7 V, above its 6 V limit. This is already listed as a residual risk in ARCHITECTURE.md.

Why it matters. The surge is short, but it can still stress the pin with repeated input transients (hot-plugging a battery, inductive cables).

Recommendation. Use R12 = 100 k, which gives a ratio of 0.104, UVLO ≈ 11.7 V and EN ≈ 6.0 V at 58 V. The internal 10.66 V UV fault still applies. Better still, add a small zener or clamp on EN once the library has one.

M7 Medium Bulk capacitor ripple and placement

Observed. C56/C57 (100 μ F / 63 V polymer, THT) sit at the left end of the bus band ($x = 93$ mm). The legs are at $x = 104.5$, 120.8 and 137 mm, so leg C is about 44 mm away. The LAYOUT note asks for them "between the half bridges".

Why it matters. The capacitor ripple current reaches roughly $0.6 \times I_{phase,rms}$, about 6 A rms at 10 A. The far legs rely on their derated 10 μ F ceramics and the band's inductance.

Recommendation. Check the ripple rating of 107AVG063MFB against about 6 A rms total at the expected temperature. Consider a third bulk cap at the leg C end, or moving one cap between legs B and C.

L1 Low Documentation mismatch

Observed. Schematic page 1 says "PCB: 2 layers, 1 oz...". The board is 4 layers (user-approved), as the README and fab notes say.

Why it matters. Reviewers and the fab order can be misled.

Recommendation. Update the root-sheet text.

L2 Low TMP235 placement

Observed. U8 is at (127.0, 126.6) mm, about 16 mm from Q5 and Q7, between the phase terminals J6 and J7.

Why it matters. It measures terminal and board temperature and lags behind the FET temperature, so over-temperature protection reacts late.

Recommendation. Move it between the low-side FETs (Q5/Q7) on the GND copper, away from the shunts, as its LAYOUT note says.

L3 Low Gate-drive tuning options

Observed. The gates are driven by IDRIVE only, with no series-resistor or RC-snubber footprints.

Why it matters. The DRV8323 power-on default (IDRIVEP 1 A / IDRIVEN 2 A) switches these MOSFETs in a few ns. That causes high dV/dt, ringing and EMI.

Recommendation. Set IDRIVE to about 100–300 mA over SPI at start-up. Optionally add 0 Ω gate-resistor and snubber footprints in Rev C.

L4 Low Hot-loop via count

Observed. C50–C55 each connect to L2 with one 0.3 mm via. The loop (cap → HS FET → LS FET → shunt → GND) is about 33 mm long and closes through L2.

Why it matters. Running directly over L2 keeps the inductance moderate, but each single via adds about 0.5 nH, and the switch-node overshoot scales with it.

Recommendation. Use 2–3 vias per cap GND pad. Optionally place the 100 nF cap next to the LS source or shunt end.

L5 Low Sensorless control only

Observed. There is no Hall, encoder or phase-voltage (BEMF) input. J2 carries 3V3, IO25, IO26 and GND.

Why it matters. Sensorless FOC with an observer is possible, but starting under load and running at low speed is harder.

Recommendation. If sensored motors are a target, add a 5-pin Hall/encoder header with RC filters in Rev C.

L6 Low Current amplifier gain

Observed. The DRV8323 CSA is at its default 20 V/V: ± 15 A gives 1.05–2.25 V (about 1.2 V of the roughly 2.3 V ADC span).

Why it matters. Resolution at 5–10 A is lower than it could be.

Recommendation. Use 40 V/V over SPI if the peak current stays ≤ 10 A (0.85–2.45 V).

L7 Low DC-bias derating

Observed. The hot-loop and VM bulk caps are C3216X6S2A106K (10 μ F / 100 V, 1206, X6S).

Why it matters. At 36 V a 1206 100 V X6S part keeps only a fraction of its nominal capacitance (check the TDK curve).

Recommendation. Use a 1210 part, or add a second cap per leg, if the measured switch-node ringing calls for it.

L8	Low	Housekeeping
Observed. There are 11 ERC warnings because the library's connector pins are typed bidirectional. 25 references are on F.Fab only. Parts sit within 3.5 mm of the top and bottom edges, so the board needs breakaway rails. Why it matters. These are all known and documented. The rails and the Fab-only references must reach the assembler. Recommendation. Fix the pin types upstream (passive). Send the assembly PDFs with the order, and add the rails when panelizing.		

5 Rule compliance (METU PowerLab checklist)

Rule	Status	Evidence
1.1 A4 + PowerLab sheet, title block	☐ Met	All 11 pages; variables filled
1.2 Hierarchical sheets, root page, global labels	☐ Met	8 sheets, Half Bridge reused 3×; root sheet has specs, block diagram, history
1.3 PowerLabKiCadLibraries only	☐ Met	Missing parts added upstream (PRs #2, #3, #5)
1.4/1.5 Wiring, grid, rail naming, symbol direction	☐ Met	Netlist checked: 79 nets, 0 shorts; rails named by domain
1.7 LAYOUT: notes	☐ Partly	Present on every critical part; several not met on the PCB (H2, M7, L2)
1.9 ERC 0 errors	☐ Met	0 errors / 11 warnings
2.0 Standard-price spec	☐ Partly	4 layers (approved surcharge); everything else standard (1 oz, 0.3 mm holes, 6/6 mil class, HASL)
2.6 DRC constraints and net classes	☐ Met	HV class 0.6 mm (IPC B2) with documented waivers in .kicad_dru
3.1 Outline, NPTH M3 holes, fixed parts	☐ Met	R1 corners, 4 × M3 NPTH, connectors on the edges
3.2 Placement, decoupling, fiducials, test points	☐ Met	Caps at their pins with vias at the pad; 3 fiducials; test points on all rails
3.3 High-current sizing / share across layers	☐ Not met	Single-layer 1 oz pours; phase lanes ≈ 8 A at ΔT 10 °C (H2)
3.3 45° routing, no stubs, clearances	☐ Met	angle_check.py, DRC
3.4 Via budget (≥ 2 per power layer change, about 1 A/via)	☐ Not met	1–2 vias at the shunt GND pads; 1 per hot-loop cap (H2, L4)
3.4 Tented vias (except EP pad side)	☐ Partly	Thermal vias also open on the bottom (M4)
3.5 Pours, islands, refill	☐ Met	Zones refilled; GND islands stitched
3.6 Silkscreen, board info, logo, QR	☐ Met	1.0 mm refs, input/phase labels, logo + QR + text variables on the bottom
3.7 Fab outputs	☐ Met	Gerber/drill zip, BOM, position file, assembly drawings

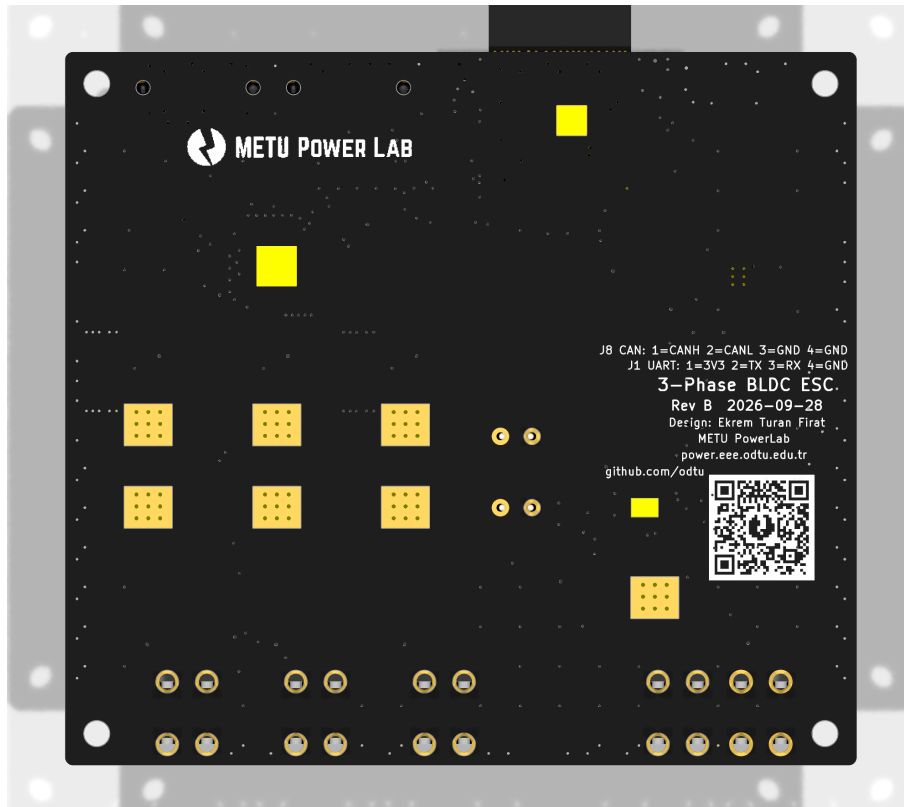
6 What the design does well

- Datasheet-driven part values with the calculations written down (eFuse IOCP/ILIM/dVdT, buck on-time and FB divider, INA240 gain choice, CAN TVS working voltage).
- The Rev A to Rev B audit trail lists every error and its fix. The netlist was checked net by net, independently of ERC.
- A solid L2 GND plane under the whole board. GND pads have their own vias, and stitching runs on a 5 mm grid and along the edges.
- HV clearance applied with a real IPC-2221B basis, with narrow and documented waivers instead of a blanket relaxation.
- Kelvin current sensing through net ties on every shunt. Gate pairs are hand-routed with their own phase returns.
- The ESP32 antenna overhangs the edge (Espressif HDG). Strapping pins were analysed (IO2/IO12 pull-downs, IO5 pull-up against the DRV8323 nSCS pull-down).
- Reproducible: the generation scripts in hardware/tools/pcb record how every placement, pour and route was made.
- Complete PCBWay package with BOM notes (wrong library distributor numbers flagged, CAN termination DNP rule).

7 Recommended actions

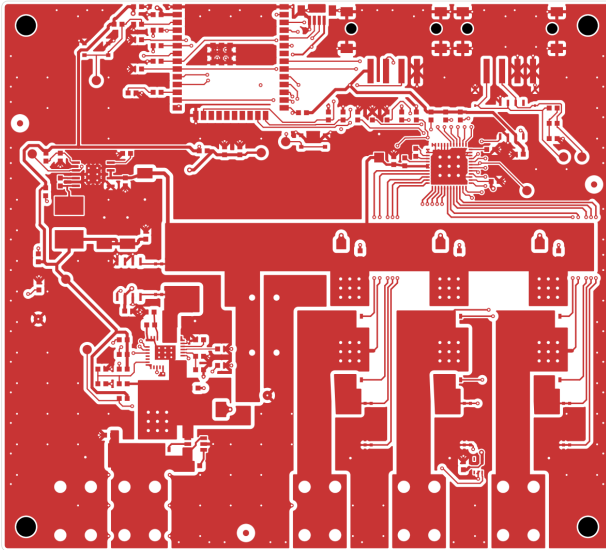
#	When	Action	Closes
1	Before motor tests	Limit braking or regeneration in firmware; run from a battery or behind a clamp; set IDRIVE low and CSA gain over SPI	H1, L3, L6
2	Before ordering assembly	Confirm TPS16890 production part and its availability; update the library fields	M3
3	Before ordering	Decide on the bottom mask openings (tent them, or keep them for an insulated heatsink); fix the root-sheet text	M4, L1
4	Rev C — power	TVS + divider on +VBUS_PROT; bus-voltage ADC on IO35; INA240 REF offset; UVLO divider R12 = 100 k	H1, M1, M2, M6
5	Rev C — copper	Copy power pours to L4/L3 with via arrays; 8–12 vias per shunt GND; widen the U3 necks; 2–3 vias per hot-loop cap	H2, L4
6	Rev C — placement	Bulk cap near leg C or between legs; TMP235 between the LS FETs; optional Hall header	M7, L2, L5
7	Bring-up	Thermal test at 5 A and 10 A (thermocouple on the FET cases); scope the switch node; check the QR scan and Gerbers	M5, L7

Appendix A — Board views

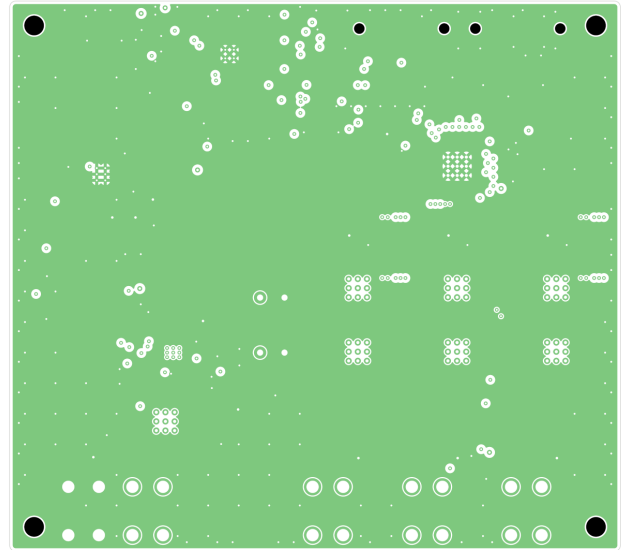


Bottom render: bare MOSFET drain pads with open vias and bare EP squares (M4); logo, board info and QR code.

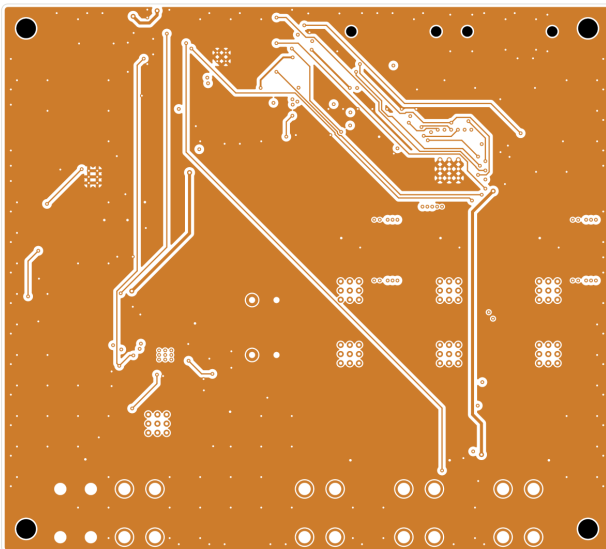
Appendix B — Copper layers (kicad-cli plots)



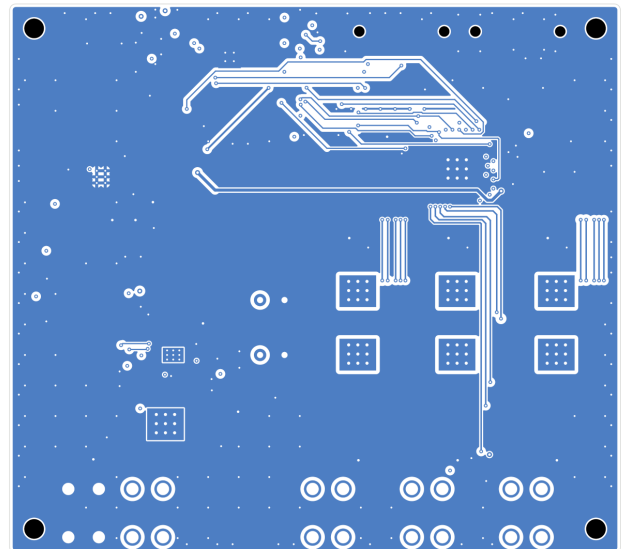
L1 (F.Cu)



L2 (In1.Cu) — solid GND



L3 (In2.Cu) — signals + GND fill



L4 (B.Cu) — signals + GND fill

Appendix C — Method and assumptions

- ERC/DRC were run with kicad-cli 10.0.5 on the committed project (zones refilled, schematic parity on). The JSON reports were parsed for counts.
- Copper areas, narrowest widths, via counts and pad mask openings were read from the filled zones and pads through the KiCad pcbnew Python API. Widths in Figures 2–4 were measured on a 1 mm grid.
- Ampacity uses IPC-2221 ($I = k \cdot \Delta T^{0.44} \cdot A^{0.725}$, $k = 0.048$ outer, 1 oz = 1.378 mil), as in the lab rules. This is conservative for short necks.
- Regeneration energy: $E = \frac{1}{2} \cdot C \cdot (V_1^2 - V_0^2)$ with $C = 200 \mu\text{F}$. Ceramic bus caps were ignored (their capacitance at 36 V is small), so the real margin is slightly better than stated.
- Loss figures are first-order estimates meant to justify measurement, not a thermal simulation.
- Datasheet limits quoted: DRV8323 VM 65 V abs max, SMBJ36CA 58.1 V clamp at 10.3 A, TPS16890 EN/UVLO 6 V abs max, and the ESP32 ADC usable range (about 0.1–0.15 V up to 0.95–2.45 V by attenuation). These are as quoted in the project's ARCHITECTURE.md and the vendor documentation. Recheck them against the current datasheet revisions before changing anything.